

WIPRO TALENTNEXT PROGRAM

PBL HANDS-ON ASSIGNMENT

Name: Munagala Deepika

Roll No: 23BQ1A4932

PBL ID: DB_262790191

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Topic: Using the Set Operators & Displaying Data from Multiple Tables Using Joins

Question 1

Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 10, 20, and 30, giving each column an appropriate heading.

Query:

```
SELECT JOB, SUM(DECODE(DEPTNO,10,SAL,0)) AS DEPT10,  
SUM(DECODE(DEPTNO,20,SAL,0)) AS DEPT20,  
SUM(DECODE(DEPTNO,30,SAL,0)) AS DEPT30, SUM(SAL) AS TOTAL FROM EMP  
GROUP BY JOB;
```

Output:

```
SQL> SELECT JOB, SUM(DECODE(DEPTNO,10,SAL,0)) AS DEPT10, SUM(DECODE(DEPTNO,20,SAL,0)) AS DEPT20, SUM(DECODE(DEPTNO,30,SAL,0)) AS DEPT  
30, SUM(SAL) AS TOTAL FROM EMP GROUP BY JOB;
```

JOB	DEPT10	DEPT20	DEPT30	TOTAL
CLERK	0	800	950	1750
SALESMAN	0	0	5600	5600
ANALYST	0	3000	0	3000
MANAGER	0	2975	2850	5825

```
SQL>
```

Question 2

Using set operator display the DEPTNO, SUM(SAL) for each dept, JOB, SUM(SAL) for each Job and Total Salary.

Query:

```
SELECT TO_CHAR(DEPTNO) AS DETAILS, SUM(SAL) AS TOTAL_SALARY FROM EMP GROUP
BY DEPTNO UNION SELECT JOB, SUM(SAL) FROM EMP GROUP BY JOB UNION SELECT
'TOTAL', SUM(SAL) FROM EMP;
```

Output:

```
SQL> SELECT TO_CHAR(DEPTNO) AS DETAILS, SUM(SAL) AS TOTAL_SALARY FROM EMP GROUP BY DEPTNO UNION SELECT JOB, SUM(SAL) FROM EMP GROUP B
Y JOB UNION SELECT 'TOTAL', SUM(SAL) FROM EMP;

DETAILS                                TOTAL_SALARY
-----                                -
20                                     6775
30                                     9400
ANALYST                               3000
CLERK                                  1750
MANAGER                               5025
SALESMAN                              5600
TOTAL                                 16175

7 rows selected.
```

Question 3

Using Set Operator display the JOB and Deptno in employees working in deptno 20,10,30 in that order.

Query:

```
SELECT JOB, DEPTNO FROM EMP WHERE DEPTNO=20 UNION SELECT JOB, DEPTNO
FROM EMP WHERE DEPTNO=10 UNION SELECT JOB, DEPTNO FROM EMP WHERE
DEPTNO=30;
```

Output:

```
SQL> SELECT JOB, DEPTNO FROM EMP WHERE DEPTNO=20 UNION ALL SELECT JOB, DEPTNO FROM EMP WHERE DEPTNO=10 UNION ALL SELECT JOB, DEPTNO F
ROM EMP WHERE DEPTNO=30;

JOB          DEPTNO
-----
CLERK        20
MANAGER      20
ANALYST      20
SALESMAN     30
SALESMAN     30
SALESMAN     30
MANAGER      30
SALESMAN     30
CLERK        30

9 rows selected.

SQL> |
```

Displaying Data from Multiple Tables Using Joins

Question 1

Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME, SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.

Query:

```
SELECT EMPNO, ENAME, SAL, DNAME, LOC FROM EMP NATURAL JOIN DEPT;
```

Output:

```
SQL> SELECT EMPNO, ENAME, SAL, DNAME, LOC FROM EMP NATURAL JOIN DEPT;
```

EMPNO	ENAME	SAL	DNAME	LOC
7000	DEEPU	800	RESEARCH	DALLAS
7001	ASHU	1600	SALES	CHICAGO
7002	BHANU	1250	SALES	CHICAGO
7003	JONES	2975	RESEARCH	DALLAS
7004	MANI	1250	SALES	CHICAGO
7698	BALU	2850	SALES	CHICAGO
7788	SCOTT	3000	RESEARCH	DALLAS
7844	TURNER	1500	SALES	CHICAGO
7900	JAMES	950	SALES	CHICAGO

9 rows selected.

```
SQL>
```

Question 2

The HR department needs a report of all employees. Write a query to display the JOB, MGR, SAL, COMM, DNAME of employees whose JOB is SALESMAN.

Query:

```
SELECT JOB, MGR, SAL, COMM, DNAME FROM EMP NATURAL JOIN DEPT WHERE  
JOB='SALESMAN';
```

Output:

```
SQL> SELECT JOB, MGR, SAL, COMM, DNAME FROM EMP NATURAL JOIN DEPT WHERE JOB='SALESMAN';
```

JOB	MGR	SAL	COMM	DNAME
SALESMAN	7698	1600	300	SALES
SALESMAN	7698	1250	500	SALES
SALESMAN	7698	1250	1400	SALES
SALESMAN	7698	1500	0	SALES

```
SQL> |
```

Question 3

The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO, and DNAME for all employees who work in DALLAS.

Query:

```
SELECT ENAME, JOB, DEPTNO, DNAME FROM EMP NATURAL JOIN DEPT WHERE LOC='DALLAS';
```

Output:

```
SQL> SELECT ENAME, JOB, DEPTNO, DNAME FROM EMP NATURAL JOIN DEPT WHERE LOC='DALLAS';
```

ENAME	JOB	DEPTNO	DNAME
DEEPU	CLERK	20	RESEARCH
JONES	MANAGER	20	RESEARCH
SCOTT	ANALYST	20	RESEARCH

```
SQL>
```

Question 4

Create a report to display employees' ename and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.

Query:

```
SELECT E.ENAME AS Employee, E.EMPNO AS "Emp#", M.ENAME AS Manager, M.EMPNO AS "Mgr#" FROM EMP E JOIN EMP M ON E.MGR=M.EMPNO;
```

Output:

```
SQL> SELECT E.ENAME AS Employee, E.EMPNO AS "Emp#", M.ENAME AS Manager, M.EMPNO AS "Mgr#" FROM EMP E JOIN EMP M ON E.MGR=M.EMPNO;
```

EMPLOYEE	Emp#	MANAGER	Mgr#
ASHU	7001	BALU	7698
BHANU	7002	BALU	7698
MANI	7004	BALU	7698
TURNER	7844	BALU	7698
JAMES	7900	BALU	7698

```
SQL> |
```

Question 5

Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.

Query:

```
SELECT E.ENAME AS Employee, E.EMPNO AS "Emp#", M.ENAME AS Manager,
M.EMPNO AS "Mgr#" FROM EMP E LEFT OUTER JOIN EMP M ON E.MGR = M.EMPNO
ORDER BY E.EMPNO;
```

Output:

```
SQL> SELECT E.ENAME AS Employee, E.EMPNO AS "Emp#", M.ENAME AS Manager, M.EMPNO AS "Mgr#"
2 FROM EMP E
3 LEFT OUTER JOIN EMP M
4 ON E.MGR = M.EMPNO
5 ORDER BY E.EMPNO;
```

EMPLOYEE	Emp#	MANAGER	Mgr#
DEEPU	7000		
ASHU	7001	BALU	7698
BHANU	7002	BALU	7698
JONES	7003	KING	7839
MANI	7004	BALU	7698
BALU	7698	KING	7839
SCOTT	7788		
KING	7839		
TURNER	7844	BALU	7698
JAMES	7900	BALU	7698

10 rows selected.

Question 6

The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.

Query:

DESC SALGRADE;

SELECT E.ENAME, E.JOB, D.DNAME, E.SAL, S.GRADE FROM EMP E, DEPT D, SALGRADE S WHERE E.DEPTNO=D.DEPTNO AND E.SAL BETWEEN S.LOSAL AND S.HISAL;

```

Command Prompt - sqlplus / X + v
SQL> CREATE TABLE SALGRADE (
  2  GRADE NUMBER(2),
  3  LOSAL NUMBER(7,2),
  4  HISAL NUMBER(7,2)
  5 );

Table created.

SQL> INSERT INTO SALGRADE VALUES (1,700,1200);

1 row created.

SQL> INSERT INTO SALGRADE VALUES (2,1201,1400);

1 row created.

SQL> INSERT INTO SALGRADE VALUES (3,1401,2000);

1 row created.

SQL> INSERT INTO SALGRADE VALUES (4,2001,3000);

1 row created.

SQL> INSERT INTO SALGRADE VALUES (5,3001,9999);

1 row created.

SQL> DESC SALGRADE;
Name                                         Null?    Type
-----
GRADE                                         NUMBER(2)

```

```

Command Prompt - sqlplus / X + v
SQL> DESC SALGRADE;
Name                                         Null?    Type
-----
GRADE                                         NUMBER(2)
LOSAL                                         NUMBER(7,2)
HISAL                                         NUMBER(7,2)

SQL> SELECT E.ENAME, E.JOB, D.DNAME, E.SAL, S.GRADE FROM EMP E, DEPT D, SALGRADE S WHERE E.DEPTNO=D.DEPTNO AND E.SAL BETWEEN S.LOSAL
AND S.HISAL;

ENAME      JOB      DNAME      SAL      GRADE
-----
KING        PRESIDENT ACCOUNTING  5000      5
DEEPU       CLERK    RESEARCH   800       1
JONES       MANAGER  RESEARCH   2975      4
SCOTT       ANALYST  RESEARCH   3000      4
JAMES       CLERK    SALES      950       1
BHANU      SALESMAN SALES     1250      2
MANI       SALESMAN SALES     1250      2
ASHU       SALESMAN SALES     1600      3
TURNER     SALESMAN SALES     1500      3
BALU       MANAGER  SALES     2850      4

10 rows selected.

```

Question 7

Display the ENAME, DNAME of all the employees. Also display those department name which do not have any employees working.

Query:

```
SELECT E.ENAME, D.DNAME FROM EMP E RIGHT OUTER JOIN DEPT D ON  
E.DEPTNO=D.DEPTNO;
```

Output:

```
SQL> SELECT E.ENAME, D.DNAME FROM EMP E RIGHT OUTER JOIN DEPT D ON E.DEPTNO=D.DEPTNO;  
  
ENAME      DNAME  
-----  
DEEPU      RESEARCH  
ASHU       SALES  
BHANU      SALES  
JONES      RESEARCH  
MANI       SALES  
BALU       SALES  
SCOTT      RESEARCH  
TURNER     SALES  
JAMES      SALES  
KING       ACCOUNTING  
           OPERATIONS  
  
11 rows selected.
```

Question 8

The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.

Query:

```
SELECT E.ENAME, E.HIREDATE, M.ENAME AS MANAGER, M.HIREDATE AS  
MANAGER_HIREDATE FROM EMP E, EMP M WHERE E.MGR=M.EMPNO AND  
E.HIREDATE<M.HIREDATE;
```

Output:

```
SQL> SELECT E.ENAME, E.HIREDATE, M.ENAME AS MANAGER, M.HIREDATE AS MANAGER_HIREDATE FROM EMP E, EMP M WHERE E.MGR=M.EMPNO AND E.HIREDATE<M.HIREDATE;  
  
ENAME      HIREDATE  MANAGER  MANAGER_H  
-----  
ASHU       20-FEB-81 BALU      01-MAY-81  
BHANU      22-FEB-81 BALU      01-MAY-81  
JONES      02-APR-81 KING      17-NOV-81  
BALU       01-MAY-81 KING      17-NOV-81  
  
SQL> |
```

Question 9

Display the EMPNO, ENAME, DNAME, LOC of those employees who are working as CLERK. Use the USING clause.

Query:

```
SELECT EMPNO, ENAME, DNAME, LOC FROM EMP JOIN DEPT USING (DEPTNO) WHERE  
JOB='CLERK';
```

Output:

```
SQL> SELECT EMPNO, ENAME, DNAME, LOC FROM EMP JOIN DEPT USING (DEPTNO) WHERE JOB='CLERK';  
  
-----  
EMPNO  ENAME      DNAME      LOC  
-----  
7000 DEEPU      RESEARCH    DALLAS  
7900 JAMES      SALES       CHICAGO  
  
SQL>
```

Question 10

Display the ENAME, SAL, MGR, DNAME of employees whose salary is more than 2000. Use the ON clause.

Query:

```
SELECT ENAME, SAL, MGR, DNAME FROM EMP JOIN DEPT ON  
EMP.DEPTNO=DEPT.DEPTNO WHERE SAL>2000;
```

Output:

```
SQL> SELECT ENAME, SAL, MGR, DNAME FROM EMP JOIN DEPT ON EMP.DEPTNO=DEPT.DEPTNO WHERE SAL>2000;  
  
ENAME      SAL      MGR  DNAME  
-----  
KING        5000      7839 ACCOUNTING  
JONES        2975      7839 RESEARCH  
SCOTT        3000      7566 RESEARCH  
BALU        2850      7839 SALES  
  
SQL> |
```

Question 11

Display the EMPNO, ENAME, JOB, DEPTNO, DNAME, LOC of employees. Use LEFT OUTER JOIN.

Query:

```
SELECT EMPNO, ENAME, JOB, EMP.DEPTNO, DNAME, LOC FROM EMP LEFT OUTER  
JOIN DEPT ON EMP.DEPTNO=DEPT.DEPTNO;
```



```
SQL> SELECT EMPNO, ENAME, JOB, EMP.DEPTNO, DNAME, LOC FROM EMP LEFT OUTER JOIN DEPT ON EMP.DEPTNO=DEPT.DEPTNO;
```

EMPNO	ENAME	JOB	DEPTNO	DNAME	LOC
7839	KING	PRESIDENT	10	ACCOUNTING	NEW YORK
7000	DEEPU	CLERK	20	RESEARCH	DALLAS
7003	JONES	MANAGER	20	RESEARCH	DALLAS
7788	SCOTT	ANALYST	20	RESEARCH	DALLAS
7001	ASHU	SALESMAN	30	SALES	CHICAGO
7002	BHANU	SALESMAN	30	SALES	CHICAGO
7004	MANI	SALESMAN	30	SALES	CHICAGO
7698	BALU	MANAGER	30	SALES	CHICAGO
7844	TURNER	SALESMAN	30	SALES	CHICAGO
7900	JAMES	CLERK	30	SALES	CHICAGO

```
10 rows selected.
```

Question 12

Display the ENAME, DNAME of employees. Use RIGHT OUTER JOIN.

Query:

```
SELECT ENAME, DNAME FROM EMP RIGHT OUTER JOIN DEPT ON
EMP.DEPTNO=DEPT.DEPTNO;
```

Output:

```
SQL> SELECT ENAME, DNAME FROM EMP RIGHT OUTER JOIN DEPT ON EMP.DEPTNO=DEPT.DEPTNO;
```

ENAME	DNAME
DEEPU	RESEARCH
ASHU	SALES
BHANU	SALES
JONES	RESEARCH
MANI	SALES
BALU	SALES
SCOTT	RESEARCH
TURNER	SALES
JAMES	SALES
KING	ACCOUNTING
	OPERATIONS

```
11 rows selected.

SQL>
```

Question 13

Display the EMPNO, DNAME, LOC of employees. Use FULL OUTER JOIN

Query:

```
SELECT EMPNO, DNAME, LOC FROM EMP FULL OUTER JOIN DEPT ON
EMP.DEPTNO=DEPT.DEPTNO;
```

Output:

```
SQL> SELECT EMPNO, DNAME, LOC FROM EMP FULL OUTER JOIN DEPT ON EMP.DEPTNO=DEPT.DEPTNO;
```

EMPNO	DNAME	LOC
7000	RESEARCH	DALLAS
7001	SALES	CHICAGO
7002	SALES	CHICAGO
7003	RESEARCH	DALLAS
7004	SALES	CHICAGO
7698	SALES	CHICAGO
7788	RESEARCH	DALLAS
7844	SALES	CHICAGO
7900	SALES	CHICAGO
7839	ACCOUNTING	NEW YORK
	OPERATIONS	BOSTON

```
11 rows selected.
```